

Fitting and Interpreting Exponential Models

Answer Key

Precalculus

Quick Answers

- | | |
|-------------------------------------|-------------------|
| 1. 320, 0.25 | 6. 1.5, 24 |
| 2. 1.5 | 7. 3.11 |
| 3. 4 | 8. 1.0188, 0.0188 |
| 4. $e^{0.405465108108164t}$, 0.405 | 9. 40, 58.77 |
| 5. 0.15, 41.76 | 10. 0.7937, 10.75 |

Common wrong answers

9. If they got 8: read the 1.08 as a constant increase of 8 units per year instead of a growth factor

1. $S(t) = 320 \cdot (1.25)^t$ thousand subscribers. At launch $t = 0$, and $b^0 = 1$, so $S(0) = 320 \cdot 1$.

$$S(0) = 320 \text{ thousand}$$

The base is the yearly multiplier. Writing $b = 1 + r$ gives $1.25 = 1 + r$, so the growth rate is $r = 1.25 - 1$, i.e. a 25 percent increase per year.

$$r = 0.25$$

2. $N(t) = a \cdot b^t$ with $N(0) = 240$ and $N(3) = 810$. From $t = 0$: $a \cdot b^0 = a = 240$. Substituting into the second point gives $240 \cdot b^3 = 810$, so $b^3 = 810/240 = 3.375$. Taking the cube root, $b = \sqrt[3]{3.375}$.

$$b = 1.5$$

Check: $240 \cdot (1.5)^3 = 240 \cdot 3.375 = 810$ cells, as recorded.

3. Solve $240 \cdot (1.5)^t = 1215$. Divide by 240: $(1.5)^t = 5.0625$. Take natural logs of both sides: $t \ln 1.5 = \ln 5.0625$, so $t = \frac{\ln 5.0625}{\ln 1.5} = \frac{1.62186}{0.405465} = 4$.

$$t = 4$$

The culture reaches 1215 cells after exactly 4 hours; check $240 \cdot (1.5)^4 = 1215$.

4. Rewrite $240 \cdot (1.5)^t$ as $240 \cdot e^{kt}$. The two forms agree when $e^k = 1.5$, i.e. $(1.5)^t = e^{(\ln 1.5)t}$ for every t , so $k = \ln 1.5$ exactly. Numerically $\ln 1.5 = 0.405465\dots$, which rounds to 0.405.

$$k = \ln 1.5 \approx 0.405$$

The continuous form is $N(t) = 240e^{0.405t}$, and $k > 0$ confirms growth.

5. $A(t) = 80 \cdot (0.85)^t$ milligrams.

(a) The base is the hourly multiplier. Writing $b = 1 - r$ gives $0.85 = 1 - r$, so $r = 1 - 0.85$: the patient loses 15 percent of the current amount each hour.

$$0.15$$

(b) Substitute $t = 4$: $A(4) = 80 \cdot (0.85)^4 = 80 \cdot 0.52200625 = 41.7605$, which rounds to 41.76.

$$A(4) \approx 41.76 \text{ mg}$$

6. $C(t) = a \cdot b^t$ with $C(1) = 36$ and $C(4) = 121.5$ square metres.

(a) Divide the second equation by the first so that a cancels: $\frac{ab^4}{ab^1} = b^3 = \frac{121.5}{36} = 3.375$, so $b = \sqrt[3]{3.375}$. $b = 1.5$

(b) Now use $C(1) = a \cdot b = 36$, so $a = 36/1.5$. $a = 24$

Check: $24 \cdot (1.5)^4 = 121.5$ square metres, matching the second measurement.

7. Doubling means $320 \cdot (1.25)^t = 2 \cdot 320$. Dividing by 320 removes it entirely: $(1.25)^t = 2$. That is why the doubling time does not depend on the initial value. Taking logs, $t = \frac{\ln 2}{\ln 1.25} = \frac{0.693147}{0.223144} = 3.1063\dots$ $t \approx 3.11$

8. The monthly factor satisfies $b^{12} = 1.25$, so $b = (1.25)^{1/12}$.

(a) $(1.25)^{1/12} = 1.018769\dots$, which rounds to 1.0188. $b \approx 1.0188$

(b) The rate is the factor minus 1: $1.018769\dots - 1 = 0.018769\dots$, about 1.88 percent per month. $r \approx 0.0188$

Note that twelve months of this factor multiply back to 1.25, while twelve times the monthly rate would overstate the yearly growth — percentages compound, they do not add.

9. $P(t) = 500 \cdot (1.08)^t$ people.

(a) $P(1) - P(0) = 500 \cdot 1.08 - 500 = 540 - 500$. 40 people

(b) $P(6) - P(5) = 500 \cdot (1.08)^6 - 500 \cdot (1.08)^5 = 793.437\dots - 734.664\dots = 58.7731\dots$, which rounds to 58.77. 58.77 people

The increases are not equal, so the growth is not a constant number of people per year. The base 1.08 multiplies the *current* population by 1.08 — an 8 percent increase — so the yearly gain itself grows. A student answering “8” has read a growth factor as though it were a constant difference.

10. $M(t) = a \cdot b^t$ milligrams with $M(0) = 12$ and $M(6) = 3$.

(a) From $t = 0$, $a = 12$. Then $12 \cdot b^6 = 3$, so $b^6 = 3/12 = 0.25$ and $b = (0.25)^{1/6} = 0.793700\dots$, which rounds to 0.7937. $b \approx 0.7937$

(b) Solve $12 \cdot b^t = 1$, i.e. $b^t = 1/12$. Taking logs, $t = \frac{\ln(1/12)}{\ln b} = \frac{-\ln 12}{-\frac{1}{6}\ln 4} = \frac{6\ln 12}{\ln 4} = \frac{6 \cdot 2.484907}{1.386294} = 10.7548\dots$ $t \approx 10.75$ hours